

RESEARCH

The 2026 AI-Engineering Reference Stack: Concrete Tooling Choices for RAG, Agents/MCP, Evaluation/Observability, Guardrails, Models, and Hosting

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Primary Question: What is a concrete, defensible default tool stack for building production LLM applications in 2026 — layer by layer, with trade-offs and a recommended pick per layer?

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CHAPTER 01

Executive Summary

CHAPTER 01

Executive Summary

The 2026 AI-engineering landscape has stabilized enough to name concrete defaults: Claude Sonnet 4.6 anchors general production use, pgvector covers most RAG deployments under 5M vectors, LangGraph dominates stateful agent orchestration, MCP has become the cross-vendor tool-integration standard, and Langfuse (now ClickHouse-backed, open-source, self-hostable) is the EU-safe observability default. Guardrails remain a defense-in-depth composition rather than a single tool, with NeMo + LLM Guard as the open-source stack and Portkey gateway adding PII redaction at the router level. This document provides the net-new layer-by-layer opinionated stack that prior researches R-00130, R-00133, and R-00148 pointed toward but did not assemble into a single decision surface.

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CHAPTER 02

Background

CHAPTER 02

Background

Three prior filed researches established the terrain but left the synthesis work unfinished:

- **R-00130** surveyed RAG/agents/evals and MCP, and concluded that "no framework dominates" at the orchestration layer — correct but not actionable.
- **R-00133** deep-dived agent evaluation and observability tooling, establishing Langfuse and Braintrust as the two main contenders without resolving the EU-data-residency dimension.
- **R-00148** covered mid-2026 model releases (Claude Opus 4.8, GPT-5.x, Gemini 3.x, DeepSeek V4), MCP standardization under the Linux Foundation, and open-weight viability, but stopped short of prescriptive layer defaults.

The gap filled here is the **concrete, opinionated per-layer recommendation**, including the EU-compliance lens that is non-negotiable for a small EU firm serving clients in regulated verticals. The EU AI Act's Annex III enforcement kicks in on **2026-08-02**, elevating compliance from aspirational to mandatory for any high-risk deployment category. (Note: per R-00148, the Digital Omnibus provisionally defers some Annex III high-risk obligations to 2027-12-02; Article 50 transparency still applies from 2026-08-02 — verify current status before relying on either date.)

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CHAPTER 03

Findings

CHAPTER 03

Findings

By mid-2026, every major provider has converged on the same rough capability tier for reasoning and code, differentiated by latency profile, context window, pricing, and constitutional/safety posture.

Layer 1 — Models [HIGH confidence]

The frontier tier (closed, API)

Anthropic Claude is the density-model holdout in an MoE world. [Claude Opus 4.8](#) (released 2026-05-28) costs \$5.00/\$25.00 per million input/output tokens with a 1M-token context. Claude Sonnet 4.6 costs \$3.00/\$15.00 per million tokens. Claude Haiku 4.5 costs \$1.00/\$5.00 per million tokens. [Batch processing reduces all rates 50%; prompt caching cuts cached-input cost 90%](#), enabling combined savings up to 95% on repetitive workloads. Claude leads on software engineering at [77.2% SWE-bench Verified](#) and supports extended autonomous sessions.

OpenAI GPT-5.x leads raw reasoning on AIME and coding throughput (LiveCodeBench). [Kimi K2 Thinking at \\$0.60/M](#) provides near-GPT-5 reasoning at a fraction of cost.

Google Gemini 2.5 Flash is the throughput and cost king among frontier tiers: [\\$0.30/\\$2.50 per million tokens, 232 tokens/second](#), making it the natural router target for high-volume commodity tasks (summarization, classification, extraction).

Mistral is the EU-native frontier option. [Mistral Large 3 costs \\$0.50/\\$1.50 per million tokens via La Plateforme](#), which runs entirely in EU data centers. Mistral Small 3.1 at \$0.10/\$0.30 covers everyday tasks. The free tier offers roughly a billion tokens/month for evaluation.

Open-weight models

The open-weight tier has closed the capability gap dramatically in 2026. [Three families are production-ready](#):

- **DeepSeek V4** (April 2026): 1M-token context, ~\$1.74/M via API or free self-host (MIT license), near-frontier benchmarks. Requires 8x H100 minimum for self-host. [Highest GPQA Diamond of any model](#).
- **Qwen 3.6** (Alibaba): Runs on a single H100 (~\$0.34/M on H100 spot, ~\$0.05/M on B200 FP4), strong on multilingual and cadence-driven workloads.
- **Llama 4 Scout/Maverick** (Meta): Apache 2.0, free to download; Scout targets single-GPU deployments, Maverick needs multi-GPU.
- **Mistral Small/Large** (open-weight versions): EU-jurisdiction training origin, Apache 2.0, deployable on-prem.

As of mid-2026, [every frontier model except Claude is mixture-of-experts](#); Anthropic argues dense models are easier to align and interpret.

Multi-model routing

Leading organizations route rather than commit. The recommended split by task type from multiple practitioner analyses:

Task type	Cheap model	Why
Classification, extraction, formatting	Gemini Flash / Haiku / Mistral Small	Commodity; frontier overkill
Code generation, extended autonomy	Claude Sonnet	SWE-bench lead
Long document reasoning, math	DeepSeek V4 / GPT-5.x	Context + reasoning
High-volume EU-data workloads	Mistral Small/Large on La Plateforme	EU residency

Smart routing achieves 30–85% cost reduction while maintaining quality, with 70/30 cheap-to-frontier splits yielding ~56–69% savings.

Router tooling: [LiteLLM](#) is the self-hostable standard (OpenAI-compatible, 100+ providers, YAML-configured routing). [Portkey](#) went Apache 2.0 in March 2026 and adds production safety (PII redaction, jailbreak detection, audit trails) at the gateway layer. Under \$2,000/month LLM spend: Portkey Free or OpenRouter. Over \$2,000/month: self-hosted LiteLLM or Portkey.

EU/GDPR model note: Only Mistral (La Plateforme, Paris) qualifies as natively EU-sovereign. Anthropic via AWS-EU Frankfurt, OpenAI Enterprise via Azure-EU Frankfurt, and Google Vertex AI europe-west qualify for data residency with Enterprise contracts (+5–20% surcharge), but remain subject to the US CLOUD Act unless contractual carve-outs are established. EU data residency means the data sits in an EU region; EU data sovereignty means no non-EU jurisdiction can compel access — these are distinct and the difference matters under Schrems II.

RECOMMENDED DEFAULT — Layer 1:

- Portfolio/senior engineer: Claude Sonnet 4.6 as primary, Gemini Flash as routing target for high-volume tasks, routed via LiteLLM or direct SDK calls.
- EU client work: Mistral Large 3 (La Plateforme) as the primary, Mistral Small 3.1 for high-volume tasks; add Claude via AWS-EU Enterprise contract for tasks where Mistral quality falls short. Self-host Llama 4 / Mistral open-weight models on EU cloud (Hetzner, OVH, Scaleway) for maximum sovereignty.

Layer 2 — RAG / Retrieval [HIGH confidence]

Vector databases

Five systems dominate production deployments. The choice hinges primarily on whether you already run PostgreSQL and on vector count:

System	Scale sweet spot	Hybrid search native	Self-host cost	Best for
pgvector 0.9	Under 5M vectors on one Postgres instance	Via pg_bm25 (ParadeDB)	Free (on existing Postgres)	Teams already on Postgres; transactional consistency needed
Qdrant	5M–hundreds of millions	Yes (sparse + dense)	\$30–50/mo self-host	Complex payload filtering (legal, financial)
LanceDB	1M+ with bulk ingestion	No native	Under \$30/mo	Embedded, serverless, larger-than-memory local datasets
Pinecone	Billions (managed, serverless)	Sparse-dense hybrid	\$70–300/mo managed	Zero DevOps teams; rapid prototyping
Weaviate	Any, but resource-heavy	Native multi-modal	\$50–100/mo self-host	Multi-modal retrieval; GraphQL query interface

[pgvector 0.9 \(early 2026\)](#) added IVFFlat improvements, sparse vector support, and speed boosts. [The vector database accounts for roughly 5–10% of RAG system quality](#); chunking strategy, embedding model, and retrieval pipeline matter far more.

Embedding models

[No single model wins everything](#). Recommended picks by scenario:

- **OpenAI text-embedding-3-large** (\$0.13/M tokens, 3072 dims): Best for existing OpenAI deployments. The practical default when you have no sovereignty constraint.
- **BGE-M3** (BAAI, open-source, 568M params, 1024 dims): [Produces dense + sparse + multi-vector ColBERT embeddings in a single forward pass](#), 100+ languages, 8192-token context. The default for teams self-hosting or needing EU data sovereignty.
- **Cohere Embed v4**: Supports 100+ languages, produces both dense and sparse in one call (simplifies hybrid search), strong enterprise retrieval. ~\$0.07/M tokens.
- **Voyage voyage-3-large** (\$0.18/M): Highest retrieval benchmark scores, best for code/technical search. High cost limits scale.
- **Gemini Embedding 2**: Best multimodal embedding (text, image, video, audio, PDF) with 32K context; relevant for document-heavy pipelines.
- Self-hosting break-even: [roughly 10–50M embeddings/month](#), after which BGE-M3 or Jina v4 GPU costs undercut hosted API pricing.

Hybrid search and reranking

[Hybrid retrieval is the minimum viable baseline for any production RAG deployment in 2026](#).

The canonical pattern:

1. BM25 retrieval top-50 (catches exact-keyword, named entity, product code matches)
2. Dense bi-encoder retrieval top-50 (catches semantic/paraphrase matches)
3. Reciprocal Rank Fusion (RRF, k=60) merging into top-100 candidates
4. Cross-encoder reranker to top-10 before passing to LLM context

[Adding a cross-encoder reranker \(Cohere Rerank v4.0 or Voyage rerank-2.5\) yields +17.2pp MRR@3 and +12.1pp Recall@5](#) over unranked hybrid retrieval. For EU-sovereign pipelines, the open-source BGE Reranker v2 runs on-prem with comparable quality.

RECOMMENDED DEFAULT — Layer 2:

- Portfolio/senior engineer: pgvector (on existing Postgres) + OpenAI text-embedding-3-large + BM25 via pg_bm25 + Cohere Rerank. Upgrade to Qdrant if crossing 5M vectors or if complex metadata filtering is needed.
- EU client work: pgvector (Supabase EU region or self-hosted Postgres) + BGE-M3 (self-hosted or Hugging Face Inference EU endpoint) + BM25 + BGE Reranker v2 (self-hosted). This stack keeps all data in-EU with zero vendor dependency for embedding or reranking.

Layer 3 — Agent / Orchestration Frameworks [MEDIUM confidence]

Note: The framework landscape is fast-moving; specific version recommendations may shift quarterly.

[Production rankings by 2026 deployment counts:](#)

1. **LangGraph** — Graph-state orchestration with explicit TypedDict state, loop support, built-in checkpointing (PostgreSQL-backed in production), and native LangSmith observability. Best for complex stateful workflows with human-in-the-loop review requirements. [Use LangGraph when the agent needs to loop, branch, resume after failure, or wait for human approval mid-execution.](#)
2. **OpenAI Agents SDK** — Lowest-friction path for GPT-centric agents. April 2026 update adds native sandboxing, sub-agents, and first-class MCP support. Best for routing-heavy workflows with clear handoff patterns.
3. **Anthropic Claude Agent SDK** — Memory API beta (February 2026). Best for Claude-native production agents; pairs naturally with Sonnet 4.6 for extended autonomous sessions.
4. **CrewAI** — Role-based multi-agent crews. Highest message-passing cost at scale, but the most intuitive mental model for non-engineers describing workflows. Best for content generation, research synthesis.
5. **AutoGen/AG2** — Reached GA February 2026 with v2 event-driven architecture. Remains strongest for multi-agent debate/critique patterns.
6. **LlamaIndex** — RAG-first with first-class agent primitives. Best when the agent's primary job is reasoning over indexed private data.
7. **Plain-code (no framework)** — Always the right choice for simple single-tool or two-tool agents; the framework abstraction tax is real.

[Framework selection should follow workflow definition, not precede it.](#) Stars measure interest, not reliability. A 40K-star framework with poor error handling fails faster than a 5K-star one with solid retry logic.

MCP (Model Context Protocol)

[MCP has achieved near-universal adoption: ~97M monthly SDK downloads, native support in Claude, ChatGPT, Gemini, Copilot, and Cursor, 10,000+ servers indexed.](#) The protocol was donated to the Linux Foundation (Agentic AI Foundation) in December 2025 with Anthropic, OpenAI, AWS, Google, Microsoft, and Cloudflare as backers. [41% of surveyed software organizations have MCP servers in limited or broad production.](#)

MCP is the correct default for tool integration when the tool needs to be accessible from multiple AI-agent frameworks or IDEs. Key caveats: [30+ CVEs were filed January–February 2026,](#) covering tool poisoning, command injection, and supply chain attacks. Production MCP deployments need gateway infrastructure for access control and audit logging.

Structured outputs

[Native structured output via constrained decoding is now supported by OpenAI, Anthropic, and Google Gemini.](#) Plain JSON mode is obsolete. Always validate with Pydantic (Python) even when the provider guarantees schema compliance — provider-side constrained decoding guarantees schema syntax, not semantic correctness.

RECOMMENDED DEFAULT — Layer 3:

- Portfolio/senior engineer: LangGraph for anything requiring loops/state/HITL; plain Python with the provider SDK (Anthropic or OpenAI) for simple single-tool agents; MCP for any tool that should be reusable across contexts.
- EU client work: Same framework choice, but prefer the Claude Agent SDK or LangGraph backed by a self-hosted PostgreSQL checkpointer on EU infrastructure; avoid MCP servers whose provenance is unclear (MCP supply chain risk is a current active concern).

Layer 4 — Evaluation & Observability [HIGH confidence]

Platform	Architecture	Pricing	EU data residency	Best for
Langfuse	Open-source (MIT), self-host on Postgres + ClickHouse	Free self-host; cloud \$59/seat	Self-host: full control; cloud: EU AWS eu-west-1	Framework-agnostic teams; EU-compliance-first
LangSmith	Closed SaaS	\$39/seat/month + \$0.50/1k traces	US-hosted (enterprise EU contract possible)	LangChain/LangGraph-native teams
Braintrust	Managed SaaS	Free to \$200/month	US-hosted	CI/CD quality gates; high scale
Arize Phoenix	Open-source, OpenTelemetry-native	Free OSS; enterprise cloud	Self-host: full control	OTel-native infra; ML-grade evaluation rigor
DeepEval	Open-source, pytest-integrated	Free	Self-host	Engineering teams with strong CI/CD; 50+ metric library
RAGAS	Open-source	Free	Self-host	Reference-free RAG metrics; custom eval infrastructure

[Langfuse was acquired by ClickHouse in January 2026](#), with the ClickHouse analytics backend now underpinning both cloud and self-hosted deployments. The project remains MIT-licensed. [Braintrust raised \\$80M Series B at \\$800M valuation in February 2026](#). LangSmith's deep advantage is zero-config tracing inside LangChain/LangGraph; the trade-off is vendor coupling and US-hosted data.

Evaluation patterns

The production evaluation loop has consolidated around:

1. **Offline golden-set evaluation:** 50–500 curated Q&A pairs with human-labeled ground truth, run on every PR.
2. **LLM-as-judge scoring:** Faithfulness, relevance, context precision/recall (RAGAS metrics); hallucination detection. Judge model should be a different provider from the production model to avoid self-evaluation bias.
3. **Production trace sampling:** 5–10% of live traces scored automatically; anomalies trigger alerts.
4. **CI/CD quality gates:** Braintrust and DeepEval (via pytest) both support blocking merges when eval scores regress below threshold.

RECOMMENDED DEFAULT — Layer 4:

- Portfolio/senior engineer: Langfuse cloud (EU region) for tracing; RAGAS or DeepEval for offline eval metrics in pytest; LLM-as-judge via Langfuse's built-in scorer pointing at Haiku/Flash as the judge model.
- EU client work: **Self-hosted Langfuse** (Postgres + ClickHouse on EU infrastructure) as the tracing and eval backbone — the only full-stack option with true EU data residency, no per-seat pricing, and MIT license. Add Arize Phoenix self-hosted alongside for adversarial eval and drift detection in regulated verticals.

Layer 5 — Guardrails & Security [MEDIUM confidence]

Note: Tool benchmarks are hardware- and benchmark-dependent; verify against your specific deployment configuration.

OWASP LLM Top 10 (v2.0, 2025–2026)

The [OWASP LLM Top 10 version 2.0](#) reflects the shift to production agentic systems with tool access:

Rank	Risk	Primary Mitigation
LLM01	Prompt injection	Input semantic scoring; structural delimiters; RAG content inspection; defense-in-depth
LLM02	Sensitive data disclosure	Output PII filtering; session isolation; data classification
LLM03	Supply chain vulnerabilities	Model provenance verification; artifact scanning
LLM04	Data/model poisoning	Document validation before ingestion; anomaly detection
LLM05	Improper output handling	Treat outputs as untrusted; parameterized queries; sandboxed execution
LLM06	Excessive agency	Least-privilege tool access; HITL for irreversible actions
LLM07	System prompt leakage	Move secrets to tool layer; monitor responses for prompt signatures
LLM08	Vector/embedding weaknesses	Access controls on vector DBs; document-level authorization
LLM09	Misinformation	Human review gates for high-stakes outputs; calibrated uncertainty
LLM10	Unbounded consumption	Output token limits; per-user rate limits; max agent step counts

Guardrail tooling

The [2026 production consensus is that guardrails are complementary, not competitive](#). The recommended layered architecture:

- **Input scanner (fast):** LLM Guard (open-source, <50ms) as a first-layer scanner for known injection patterns, PII detection, and toxicity. [Portkey gateway](#) (Apache 2.0 since March 2026) adds these at the routing layer without application-code changes.
- **Dialog rails:** [NVIDIA NeMo Guardrails](#) (Apache 2.0, Colang DSL, sub-50ms on GPU) for programmable conversation flow constraints; required for regulated environments where compliance reviewers need a readable allowed-intents document.
- **Output validation:** [Guardrails AI](#) (60+ Python validators, 50–200ms) for structured output enforcement, PII stripping, schema validation, and JSON compliance; pairs naturally with Pydantic.
- **Safety classifier:** [Llama Guard 4 / Llama Guard 3](#) as an optional final input/output content classifier; roughly one-third the false-positive rate of GPT-4 on Meta's benchmark. Only warranted for high-risk surfaces.

[No single tool covers all 10 OWASP risks](#). The silent structural defense that matters most is **least-privilege tool design** (LLM06): agents should only have access to the operations they actually need, with irreversible actions (delete, send, charge) gated behind human approval.

RECOMMENDED DEFAULT — Layer 5:

- Portfolio/senior engineer: Pydantic output validation (always), LLM Guard or Portkey for input scanning, least-privilege tool design. Skip the dialog DSL unless you have compliance reviewers.
- EU client work: NeMo Guardrails (on-prem, dialog policy as readable Colang document for

compliance review) + Guardrails AI (output enforcement) + Llama Guard 4 for content moderation on high-risk surfaces. All three run entirely on EU infrastructure; the Colang dialogue rules can serve as part of the required EU AI Act technical documentation of system intent.

Layer 6 — Serving / Hosting & Inference [HIGH confidence]

Self-hosted inference

[TGI entered maintenance mode December 2025](#). Do not start new TGI deployments. The current field:

Engine	Throughput (H100)	Best for	Notes
vLLM	~12,500 tok/s	Production multi-user serving; GPU-efficient batch inference	PagedAttention; OpenAI-compatible API; de facto standard
SGLang	~16,200 tok/s	Multi-turn/agentive workloads; highest throughput	~29% throughput advantage over vLLM; growing fast
Ollama	~41 tok/s (single-user)	Local dev; prototyping; single-user	~19x slower than vLLM at peak concurrent load
TGI	—	Do not use for new deployments	Maintenance mode; HF contributing to vLLM

[The throughput gap between Ollama and vLLM is ~19x at peak concurrent load](#). Ollama's concurrency model cannot serve production traffic to many concurrent users; it is the right tool only for solo local development.

EU cloud infrastructure for self-hosting

- **OVHcloud** (French): GPU instances (H100, A100), EU data centers, ISO 27001, GDPR-native.
- **Hetzner** (German): Most cost-competitive EU GPU hosting; H100 clusters available.
- **Scaleway** (French): Strong EU privacy positioning; GPU H100 instances.
- **Mistral La Plateforme**: For teams that don't want to operate their own inference, this is the only managed frontier-model API with native EU hosting at competitive pricing.

Cost levers

1. **Prompt caching** (provider-native): 90% discount on cached input tokens (Anthropic, Mistral, Google).
2. **Semantic caching** (application-level): [Redis LangCache or GPTCache](#); [40–80% cost reduction on repetitive queries](#). Effective when >30% of queries are semantically similar.
3. **Multi-model routing**: Route 70–80% of traffic to cheap models. [40–85% total cost reduction achievable](#).

Stacking all three can achieve 60–80% total cost reduction in optimized production deployments.

RECOMMENDED DEFAULT — Layer 6:

- Portfolio/senior engineer: Anthropic/OpenAI API directly for prototyping; add LiteLLM gateway when crossing \$500/month in API spend to enable routing and fallback; add semantic caching (Redis) when query repetition is visible in traces.
- EU client work: Mistral La Plateforme for managed frontier inference (EU-native); OVHcloud or Hetzner H100 cluster running vLLM for self-hosted open-weight models (Llama 4, Mistral open-weight); LiteLLM as the unified EU-routing gateway fronting both.

Layer 7 — App / Glue Layer for Small Teams [MEDIUM confidence]

Python backend stack

- **FastAPI** — production standard for LLM API backends: async-native, OpenAPI docs auto-generated, SSE streaming built-in via `StreamingResponse`, Pydantic v2 validation throughout.
- **uv** — replaces pip/Poetry as the package manager: [significantly faster than pip; the modern project setup standard in 2026](#).
- **Ruff** — linting and formatting (replaces flake8 + black + isort in one tool).
- **Pydantic v2** — for all data models, regardless of what else you use.
- **Docker** — containerization with multi-stage builds.

For LLM streaming: FastAPI SSE via `StreamingResponse` with an `async for` generator over the provider SDK's streaming iterator is the standard pattern. (This project, iw-ai-core, already uses this approach in its dashboard layer.)

Rapid productization stack (small firm / portfolio)

- **Supabase**: [Postgres + auth + storage + pgvector in one managed service](#). EU region (AWS eu-west-1) available.
- **Vercel**: Frontend hosting (Next.js/React). [FastAPI can deploy to Vercel via Python runtime](#), though Render or Railway are simpler for persistent FastAPI backends.
- **Stripe**: Payment integration. The Supabase/Stripe/Vercel combination has become a recognized "SaaS boilerplate" pattern in 2025–2026.
- **Railway** or **Render**: Better than Vercel for FastAPI with background tasks or long-running SSE connections.

For EU-compliance-first productization: **Supabase EU region** + FastAPI on OVHcloud/Hetzner for inference-heavy endpoints, fronted by Cloudflare (EU workers for edge filtering).

RECOMMENDED DEFAULT — Layer 7:

- Portfolio/senior engineer: FastAPI + uv + Ruff + Pydantic v2 + Docker; Supabase for the

data/auth layer; Vercel for the frontend; Railway for the FastAPI backend if you need persistent connections.

- EU client work: FastAPI + uv + Docker on EU cloud (OVHcloud or Hetzner); Supabase EU region (or self-hosted Postgres); self-hosted Langfuse for observability on the same EU cluster. Avoid Vercel for EU-sensitive data processing (US CDN edge by default); use Cloudflare with EU Data Localization Suite if edge caching is needed.

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CHAPTER 04

Recommendations

CHAPTER 04

Recommendations

The following table consolidates a single reference stack. All picks are defensible defaults, not absolute rules; the per-layer guidance notes when to deviate.

Layer	Recommended Default (Portfolio / Senior Engineer)	Recommended Default (EU Client Work)	Key Justification
1 — Models	Claude Sonnet 4.6 (primary) + Gemini Flash (volume routing) via LiteLLM	Mistral Large 3 on La Plateforme (primary) + Mistral Small 3.1 (volume) + Claude via AWS-EU Enterprise for gap tasks	Sonnet leads SWE-bench; Flash minimizes cost. Mistral is the only EU-native frontier option.
2 — Vector DB	pgvector (<5M vectors) → Qdrant when >5M or complex filtering	pgvector (Supabase EU / self-hosted) → Qdrant self-hosted (EU) for scale	pgvector eliminates infra overhead for most use cases; Qdrant scales without managed-cloud lock-in.
2 — Embeddings	OpenAI text-embedding-3-large	BGE-M3 self-hosted (or HF Inference EU endpoint)	BGE-M3 produces dense + sparse + ColBERT in one pass; EU sovereign; no per-token cost.
2 — Retrieval	Hybrid (BM25 + dense) + Cohere Rerank v4	Hybrid + BGE Reranker v2 (self-hosted)	Hybrid + reranker is the baseline; BGE keeps EU data on-prem.
3 — Agents	LangGraph (stateful/HITL) or plain SDK; MCP for tools	Same; prefer Claude Agent SDK; validate all MCP server provenance	LangGraph is the stateful standard; plain code for simple cases.
4 — Observability	Langfuse cloud (EU) + DeepEval CI gates	Self-hosted Langfuse (Postgres + ClickHouse, EU) + Arize Phoenix	Only self-hosted Langfuse gives EU residency + zero per-seat cost + OSS auditability.
5 — Guardrails	Pydantic validation + LLM Guard (input) + least-privilege tools	NeMo Guardrails + Guardrails AI + Llama Guard 4 + Portkey gateway	Defense-in-depth; NeMo's Colang doubles as compliance documentation.
6 — Serving	API → LiteLLM at \$500+/mo; Redis semantic cache for repetitive queries	Mistral La Plateforme + vLLM on OVHcloud/Hetzner H100	vLLM is the self-host standard (TGI is EOL). EU cloud keeps data sovereign.
7 — App layer	FastAPI + uv + Pydantic v2 + Docker; Supabase; Railway; Vercel	FastAPI + uv + Docker on EU cloud; Supabase EU; Cloudflare EU Data Localization	uv + FastAPI is the 2026 Python standard; avoid US-default CDNs for EU-sensitive data.

Cost profile — portfolio project (~100K LLM calls/month): Claude Sonnet \$150–300/mo, Supabase Pro \$25/mo, Langfuse cloud free tier, Railway \$20/mo → ~\$200–350/mo before routing optimization.

Cost profile — EU client deployment (same volume, fully EU-sovereign): Mistral Large 3 \$150–300/mo, Hetzner Postgres ~\$30/mo, self-hosted Langfuse + ClickHouse ~\$60/mo hardware, vLLM on shared OVH GPU node \$200–400/mo → ~\$450–800/mo for full EU sovereignty.

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CHAPTER 05

Limitations

CHAPTER 05

Limitations

Fast-moving landscape: Model names, benchmark scores, and pricing change quarterly. Specific version numbers (Claude Sonnet 4.6, Opus 4.8, DeepSeek V4, Qwen 3.6) are accurate as of 2026-06-23 and will shift — note that R-00148 already records newer Anthropic releases (Opus 4.8, Fable 5/Mythos 5); re-validate model-name-specific claims after September 2026.

Open-weight self-hosting complexity: Self-hosting vLLM + open-weight models on EU cloud is production-viable but requires GPU infrastructure expertise. The economics are favorable at scale but the operational overhead is real for a team of 1–3.

MCP security posture: The 2026 CVE history (30+ CVEs Jan–Feb 2026) means production MCP deployments require gateway enforcement and provenance checking that are not yet standardized.

EU AI Act scope uncertainty: The August 2026 / December 2027 enforcement dates for Annex III high-risk systems introduce obligations whose practical interpretation by national authorities is still being established. Guardrail recommendations here are informed by the Act's text but are not legal advice.

Benchmarks as proxies: SWE-bench, GPQA Diamond, and similar scores measure specific task distributions. Maintain an internal golden-set eval (Layer 4) reflecting your actual task distribution.

No LlamaIndex deep-dive: LlamaIndex's agent primitives and data connectors are production-mature for RAG-first workloads; this document foregrounds LangGraph for broader stateful-workflow applicability but LlamaIndex deserves consideration when the primary use case is document retrieval with LLM reasoning.

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CHAPTER 06

Sources

CHAPTER 06

Sources

Cross-referenced prior research (IW AI Core DB): R-00130

(screened-for stack), R-00133 (eval/observability tooling), R-00148 (mid-2026 models, MCP, open-weight, pricing), R-00149 (portfolio build spec).

#	Title	Credibility	URL
1	AI Model Benchmarks 2026: GPT vs Claude vs Gemini	Medium (benchmark aggregator)	https://teamai.com/blog/large-language-models-llms/the-2026-ai-frontier-model-war-2/
2	AI Leaderboard 2026: 300+ Models	High (live leaderboard)	https://llm-stats.com/
3	Anthropic Claude API Pricing	High (official)	https://platform.claude.com/docs/en/about-claude/pricing
4	Anthropic API Pricing 2026 Guide (Finout)	Medium	https://www.finout.io/blog/anthropic-api-pricing
5	Mistral AI Pricing (official)	High (official)	https://mistral.ai/pricing/
6	Best Open-Source LLMs May 2026 (Web3AI)	Medium	https://www.web3aiblog.com/blog/best-open-source-llms-llama-4-qwen-3-deepseek-v3-mistral-large-3-may-2026
7	LLM Stats — Model Updates	Medium	https://llm-stats.com/llm-updates
8	MoE Architecture: GPT/Claude/DeepSeek/Qwen	Medium	https://www.digitalapplied.com/blog/moe-architecture-comparison-gpt-claude-deepseek-qwen
9	LLM Model Routing 2026 Guide	Medium	https://www.digitalapplied.com/blog/llm-model-routing-2026-cost-quality-optimization-engineering-guide
10	OpenRouter vs LiteLLM vs Portkey 2026	Medium	https://toolhalla.ai/blog/openrouter-vs-litellm-vs-portkey-2026
11	Portkey vs LiteLLM vs OpenRouter (PkgPulse)	Medium	https://www.pkgpulse.com/guides/portkey-vs-litellm-vs-openrouter-llm-gateway-2026
12	EU Data Residency for AI (MateIT)	High (EU-law-informed)	https://www.mateit.de/en/blog/dsgvo-ki-eu-hosting/
13	Vector Database Comparison 2026 (4xxi)	High	https://4xxi.com/articles/vector-database-comparison/
14	Vector DB Benchmarks 2026 (Callsphere)	High (benchmark)	https://callsphere.ai/blog/vector-database-benchmarks-2026-pgvector-qdrant-weaviate-milvus-lancedb
15	Best Embedding Models for RAG 2026 (Milvus)	High	https://milvus.io/blog/choose-embedding-model-rag-2026.md
16	Best Embedding Models 2026 (PEC)	Medium	https://pecollective.com/tools/best-embedding-models
17	Hybrid Search: BM25 + Vector + Reranking 2026	Medium	https://www.digitalapplied.com/blog/hybrid-search-bm25-vector-reranking-reference-2026
18	Reranking & Cross-Encoders for RAG	Medium	https://localaimaster.com/blog/reranking-cross-encoders-guide
19	Best AI Agent Frameworks 2026 (AliceLabs)	Medium	https://alicelabs.ai/en/insights/best-ai-agent-frameworks-2026
20	AI Agent Frameworks (Arsum)	Medium	https://arsum.com/blog/posts/ai-agent-frameworks/
21	LangGraph State Management & Recovery	Medium	https://eastondev.com/blog/en/posts/ai/20260424-langgraph-agent-architecture/
22	MCP Adoption Statistics 2026	Medium	https://www.digitalapplied.com/blog/mcp-adoption-statistics-2026-model-context-protocol

23	The MCP Ecosystem in 2026 (ChatForest)	Medium	https://chatforest.com/guides/mcp-ecosystem-2026-state-of-the-standard/
24	ClickHouse Acquires Langfuse (official)	High (primary)	https://clickhouse.com/blog/clickhouse-acquires-langfuse-open-source-llm-observability
25	Langfuse Self-Hosting (official docs)	High (official)	https://langfuse.com/self-hosting
26	Best RAG Evaluation Tools 2026 (Braintrust)	Medium (vendor)	https://www.braintrust.dev/articles/best-rag-evaluation-tools
27	DeepEval — Evaluate RAG in CI/CD (Confident AI)	High (tooling docs)	https://www.confident-ai.com/blog/how-to-evaluate-rag-applications-in-ci-cd-pipelines-with-deepeval
28	RAGAS Paper (arXiv 2309.15217)	High (research paper)	https://arxiv.org/abs/2309.15217
29	OWASP LLM Top 10 (2026) — Repello	High	https://repello.ai/blog/owasp-llm-top-10-2026
30	AI Guardrails Compared: NeMo/Guardrails AI/Llama Guard	Medium	https://particula.tech/blog/ai-guardrails-compared-nemo-guardrails-ai-llama-guard
31	Best AI Agent Security & Guardrails Tools 2026 (DEV)	Medium	https://dev.to/agdex_ai/best-ai-agent-security-guardrails-tools-in-2026-llm-guard-vs-nemo-vs-guardrails-ai-5e5d
32	Essential Guide to LLM Guardrails (Medium)	Medium	https://medium.com/data-science-collective/essential-guide-to-llm-guardrails-llama-guard-nemo-d16ebb7cbe82
33	vLLM vs TGI vs Ollama (Birjob)	High (benchmark)	https://www.birjob.com/blog/vllm-tgi-ollama-self-hosting-llms
34	vLLM vs Ollama vs LM Studio 2026 (Codersera)	High (benchmark)	https://codersera.com/blog/vllm-vs-ollama-vs-lm-studio-production-2026/
35	EU AI Act Compliance Requirements 2026 (Secure Privacy)	High	https://secureprivacy.ai/blog/eu-ai-act-2026-compliance
36	Semantic Caching for AI Agents (BuildMVPFast)	Medium	https://www.buildmvpfast.com/blog/semantic-caching-ai-agents-cost-optimization
37	FastAPI + Supabase Template for LLM SaaS	Medium	https://euclideanai.substack.com/p/fastapi-supabase-template-for-llm
38	LLM Deployment with FastAPI + Docker + uv 2026	Medium	https://www.pyinns.com/python/llm-and-generative-ai/llm-deployment-fastapi-docker-uv-python-2026-complete-guide-best-practices
39	Deploy FastAPI on Vercel (official docs)	High (official)	https://vercel.com/docs/frameworks/backend/fastapi
40	LLM Structured Outputs in 2026 (DEV)	Medium	https://dev.to/pocket_tools/llm-structured-output-in-2026-stop-parsing-json-with-regex-and-do-it-right-34pk